

Lab: Explore a Simple Generative Tool

Estimated time needed: 30 minutes

Overview

Generative AI models have revolutionized how you interact with technology, enabling you to create new content, generate realistic images, and translate languages with remarkable accuracy.

In this lab, you will gain hands-on experience with a simple generative AI tool, DataRobot, exploring its capabilities and applications.

Learning Objectives

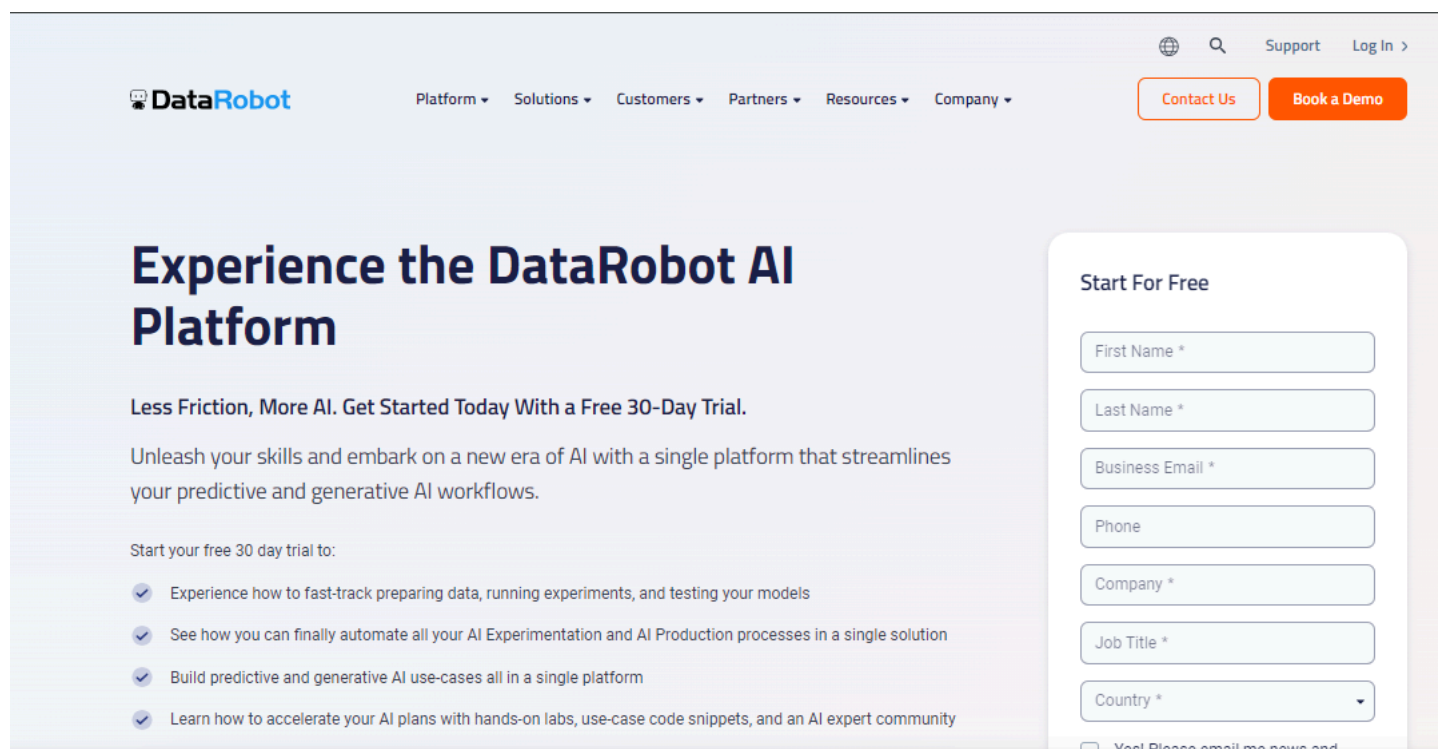
After completing this lab, you will be able to:

- Sign up in DataRobot
- Add a data set to the use case
- Work on model building

Task 1: Sign-up in DataRobot

Step 1: Click www.datarobot.com

Step 2: Fill in the required information under the "Start for free" section and create an account.



The screenshot shows the DataRobot website's homepage. The header includes the DataRobot logo, navigation links (Platform, Solutions, Customers, Partners, Resources, Company), and buttons for 'Contact Us' and 'Book a Demo'. The main content area features the headline 'Experience the DataRobot AI Platform' and a sub-headline 'Less Friction, More AI. Get Started Today With a Free 30-Day Trial.' Below this, a list of benefits is provided, each preceded by a checkmark icon. On the right side, there is a 'Start For Free' sign-up form with input fields for First Name, Last Name, Business Email, Phone, Company, Job Title, and Country. At the bottom of the form, there is a checkbox labeled 'Yes! Please email me news and'.

DataRobot

Platform Solutions Customers Partners Resources Company

Contact Us Book a Demo

Experience the DataRobot AI Platform

Less Friction, More AI. Get Started Today With a Free 30-Day Trial.

Unleash your skills and embark on a new era of AI with a single platform that streamlines your predictive and generative AI workflows.

Start your free 30 day trial to:

- ✓ Experience how to fast-track preparing data, running experiments, and testing your models
- ✓ See how you can finally automate all your AI Experimentation and AI Production processes in a single solution
- ✓ Build predictive and generative AI use-cases all in a single platform
- ✓ Learn how to accelerate your AI plans with hands-on labs, use-case code snippets, and an AI expert community

Start For Free

First Name *

Last Name *

Business Email *

Phone

Company *

Job Title *

Country *

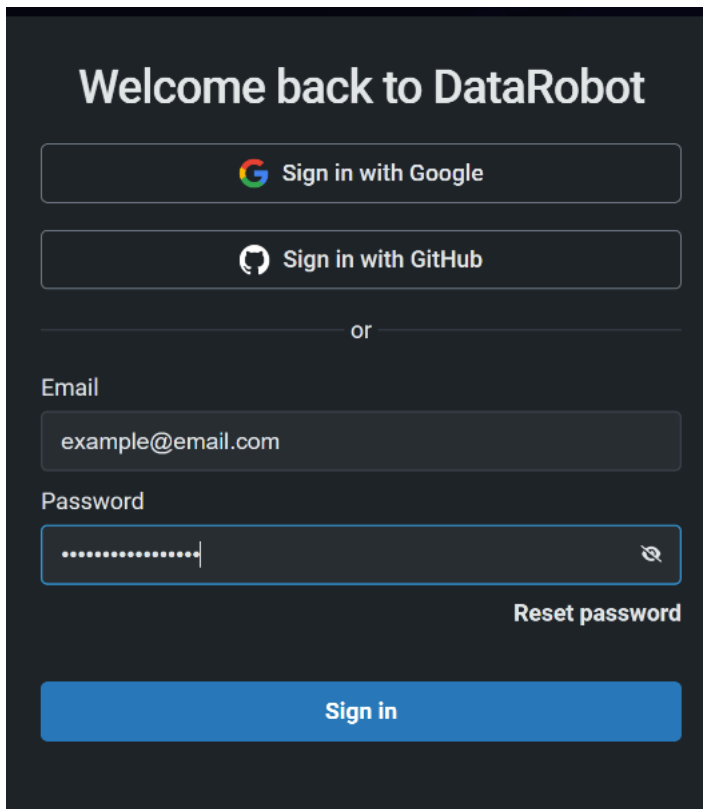
☐ Yes! Please email me news and

Note: To access the DataRobot platform, you must sign up using a work email address. If you do not have a relevant work email, an alternative is to create a GitHub account using your Gmail address. Once registered, you can log in to DataRobot using your GitHub credentials.

For step-by-step guidance on creating a GitHub account, please refer to the following link:

[GitHub Account Setup Guide](#)

Step 4: A new window will open; select the relevant option for signing up.



Welcome back to DataRobot

Sign in with Google

Sign in with GitHub

or

Email

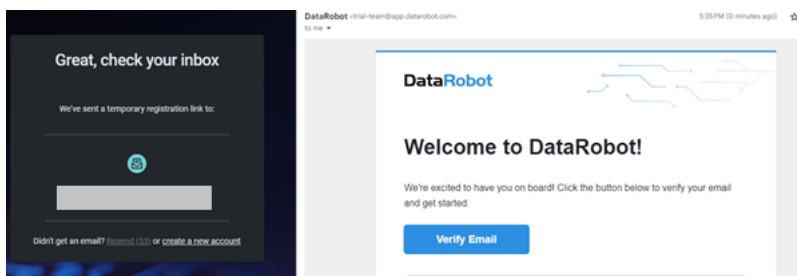
example@email.com

Password

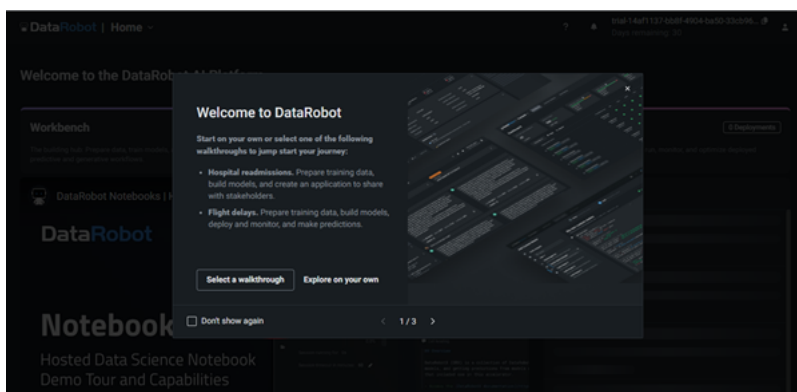
Reset password

Sign in

Step 5: Confirm your email by clicking **Verify Email** in your inbox.

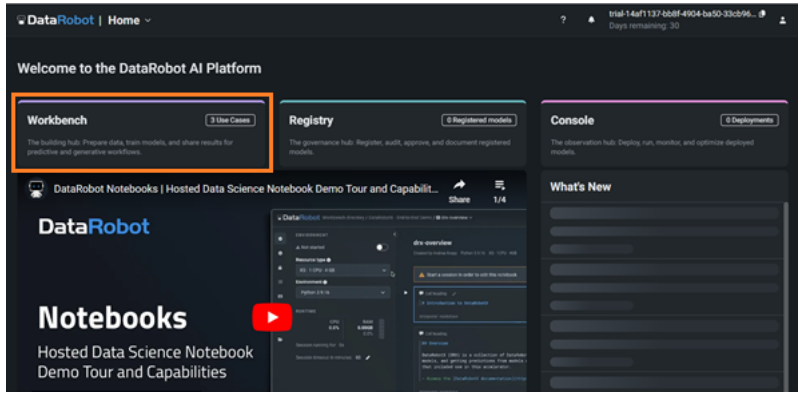


Step 6: Sign up and start your first experience of using the Generative AI tool.
The dashboard will look like the image below. You may like to familiarize yourself with the application by clicking **Select a walkthrough**.

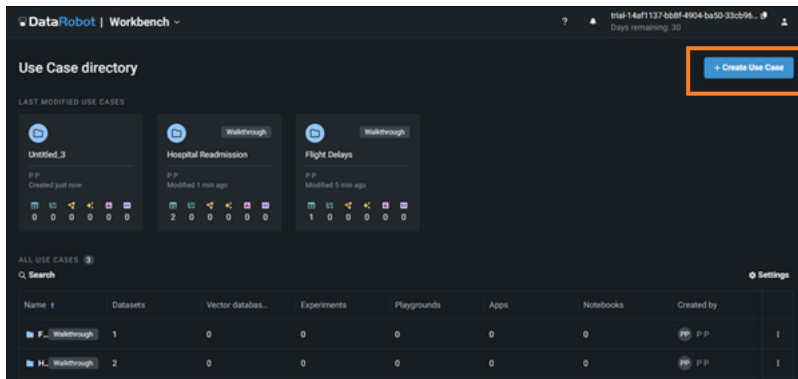


Task 2: Add a data set

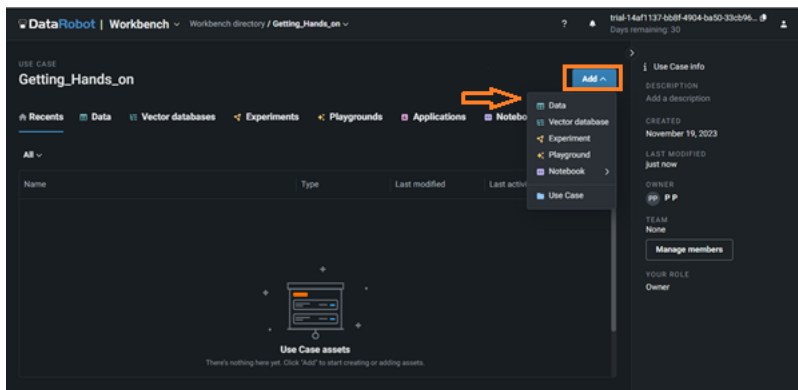
Step 7: The dashboard will appear shortly, and your screen will look as shown below. Click **Workbench**.



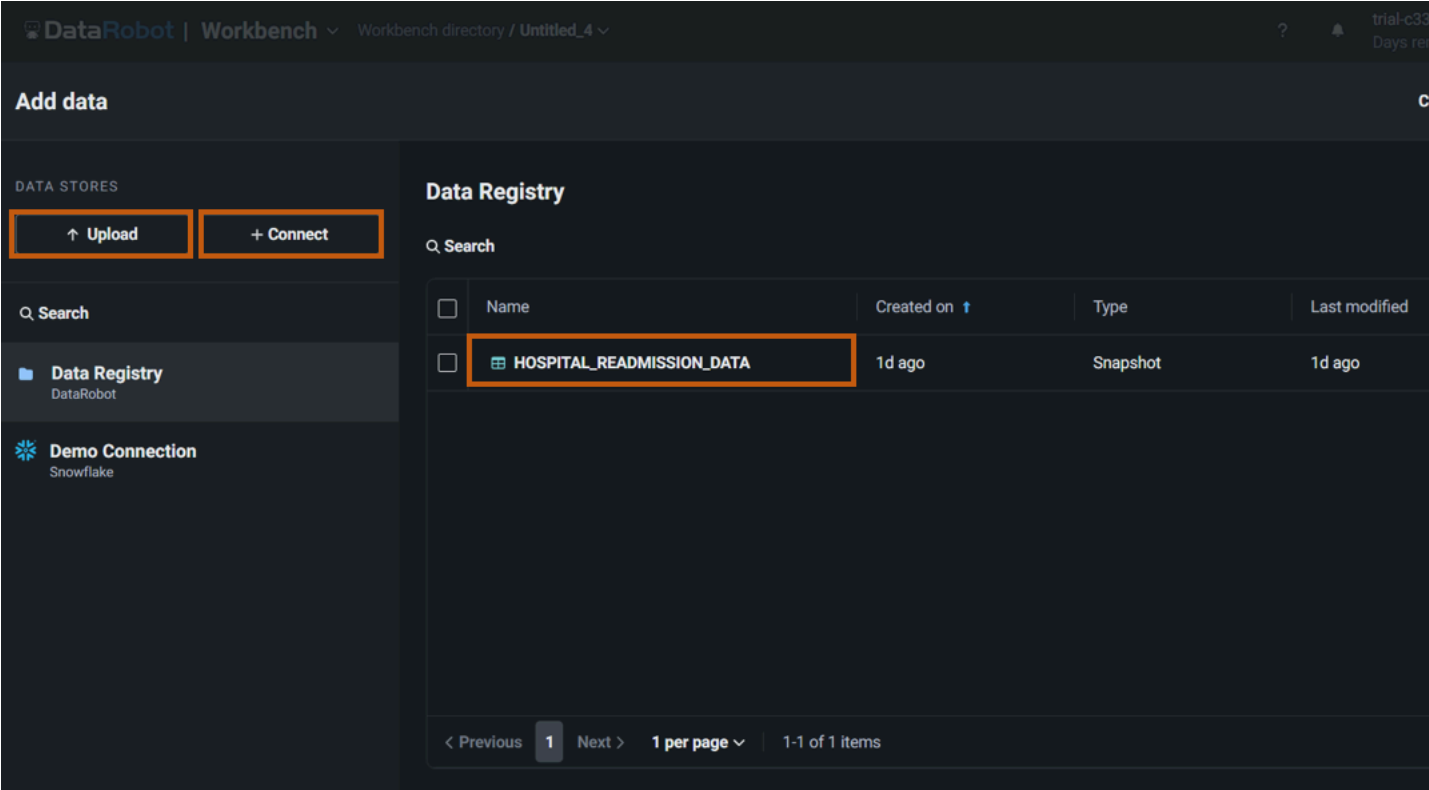
Step 8: Click **Create Use Case**.



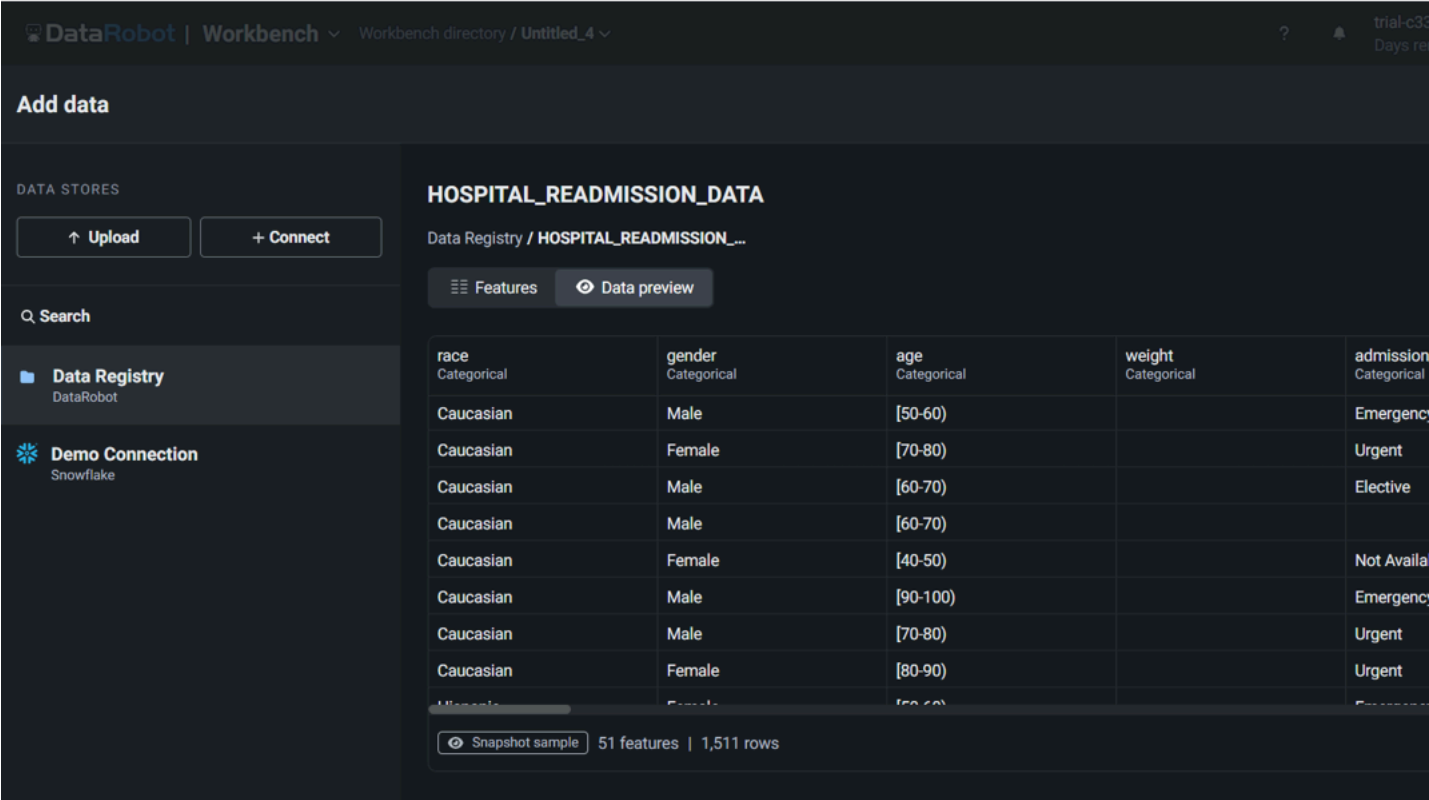
Step 9: Click **Add** and **Data** to include the data set in your use case.



Step 10: **Upload** your data set or **Connect** to the data source; however, for this lab, you can select an in-built sample data set *HOSPITAL_READMISSION_DATA*.



Step 11: Once you select the data set, you can see a preview of it. You can also view the data set's features, as shown below. Click **Add to Use Case**.



Step 12: After you add the data set to the use case, the workbench will appear as shown below. You can click the data set to see the feature insights.

DataRobot | Workbench Workbench directory / Untitled_4

< Use Case directory

USE CASE
Untitled_4 Add Data

All Data 1 Vector databases Experiments Playgrounds Applications

Type Search Settings

Name	Created By	Last Modified	Type	Source	Rows
HOSPITAL_READMISSION_DATA	PV Pratiksha V.	now	Snapshot	Snowflake	10000

< Previous 1 Next > 1 per page 1-1 of 1 items

Step 13: Explore the **All Features** menu to display specific features.

DataRobot | Workbench Workbench directory / Untitled_4 / **HOSPITAL_READMISSION_DATA**

HOSPITAL_READMISSION_DATA Jul 29th, 2024 10:39 AM Snapshot Data actions

Data preview **Features** Feature lists

☒ Show insights Show features from: All Features ^ + Create feature list

Search

DATAROBOT FEATURE LISTS

- All Features 51
- Informative Features 40
- Raw Features 51

race	age	weight	admission_type_id	discharge
Caucasian	[70-80]	==Missing==	Emergency	Discharge
AfricanAmerican	[60-70]	[75-100]	Urgent	Discharge
Other	Other	Other	Other	Other
Caucasian	[50-60]		Emergency	Discharge
Caucasian	[70-80]		Urgent	Discharge

Snapshot sample 51 features | 1,511 rows

Task 3: Work on Data Modeling

Step 14: Click **Start**. You will have options **Modelling** and **Start wrangling**. You can try data wrangling if you want to. For this lab, you will work on model building. Click **Start** and select **Modelling**. It will take a while to prepare a data set for modelling.

DataRobot | Workbench Workbench directory / Untitled_4 / HOSPITAL_READMISSION_DATA

HOSPITAL_READMISSION_DATA Jul 29th, 2024 10:39 AM Snapshot Data actions ^

Data preview Features Feature lists

Show insights Show features from: All Features + Create feature list

race	gender	age	weight	admission
Categorical	Categorical	Categorical	Categorical	Categorical
Caucasian	Female	[70-80]	==Missing==	Emergency
AfricanAmerican	Male	[60-70]	[75-100]	Urgent
Other		Other	Other	Other

Snapshot sample 51 features | 1,511 rows

Data actions menu:

- Start wrangling
- Start modeling**
- Start feature discovery
- Download dataset
- Remove dataset

Step 15: Once done, you need to select the **Target feature**. Select **readmitted** as your target feature.

DataRobot | Workbench Workbench directory / Untitled_4

Set up new experiment Dataset **Target** Additional settings

Target feature
Select the feature to make predictions on.

readmitted

	Uniq...	Missi...	Mean	Std Dev		
7	9592	-	-			
troglitazone	35	Categorical	1	0	-	-
tolbutamide	30	Categorical	2	0	-	-
tolazamide	36	Categorical	2	0	-	-
time_in_hospital	7	Numeric	14	0	4.43	3.021

Experiment
HOSPITAL_07-30 11:23

Dataset
Name

Rows
Features

Target
No target selected

Step 16: The workbench screen will be displayed as shown below. Click **Next**.

DataRobot | Workbench

Workbench directory / Untitled_4

trial-c3398et
Days remain

Set up new experiment

Dataset

Target

Additional settings

Target feature

Select the feature to make predictions on.

readmitted

Target type: Binary classification

Positive class: ☐ 0 ☒ 1

Modeling mode

Set the mode used for selecting which blueprints to build when training models.

Quick Autopilot

Optimization metric

Set the metric used when training models to evaluate and optimize accuracy.

LogLoss (Accuracy) Recommended

Number of rows

6000
5000
4000
3000
2000
1000
0

FalseTrue

Values of readmitted

Experiment

HOSPITAL_P
07-30 11:23

Dataset

Name

Rows

Features

Target

Feature

Target type

Positive class

Modeling mo

Optimization

Training featu

Partitioning

Step 17: You can modify the model setting in **Additional Settings**; once done, click **Next** and then click **Start modelling**.

DataRobot | Workbench

Workbench directory / Untitled_4

trial-c3398et
Days remain

Set up new experiment

Dataset

Target

Additional settings

Data partitioning

Time series modeling Preview

Additional settings

Partitioning method

Select the method for assigning rows to partitions when training models.

Stratified sampling
Rows are assigned to ensure similar target distribution across each partition.

Validation type

☒ Cross-validation
Trains models on a specified number of folds, maximizing data use but also increasing run time.

☐ Training-validation-holdout
Splits data into three partitions: trains models on the training set, assess performance on the validation set, and evaluates the model on unseen data in the holdout set.

Cross-validation folds

Enter a value from 2 - 50.

Holdout percentage

Set the subset of data that is unavailable during training and validation. Enter a value

Experiment

HOSPITAL_P
07-30 11:23

Dataset

Name

Rows

Features

Target

Feature

Target type

Positive class

Modeling mo

Optimization

Training featu

Partitioning

Step 18: Building models will take a while.

The screenshot shows the DataRobot Workbench interface. The top navigation bar includes the DataRobot logo, 'Workbench', and a breadcrumb trail: 'Workbench directory / Untitled_4 / HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57'. The left sidebar has tabs for 'Experiment' and 'Comparison'. Under 'Experiment', there are options to 'View experiment info', 'Filter' (0), 'Validation' (LogLoss), and a 'Search' bar. The 'MODELS' section shows '0/0' models. A large circular progress indicator in the center of the main area indicates 'Models are building...'. The left sidebar also lists two models in progress: 'Keras Slim Residual Neural Network Classifier using...' and 'Elastic-Net Classifier (L2 / Binomial Deviance)', both marked as 'Building...'.

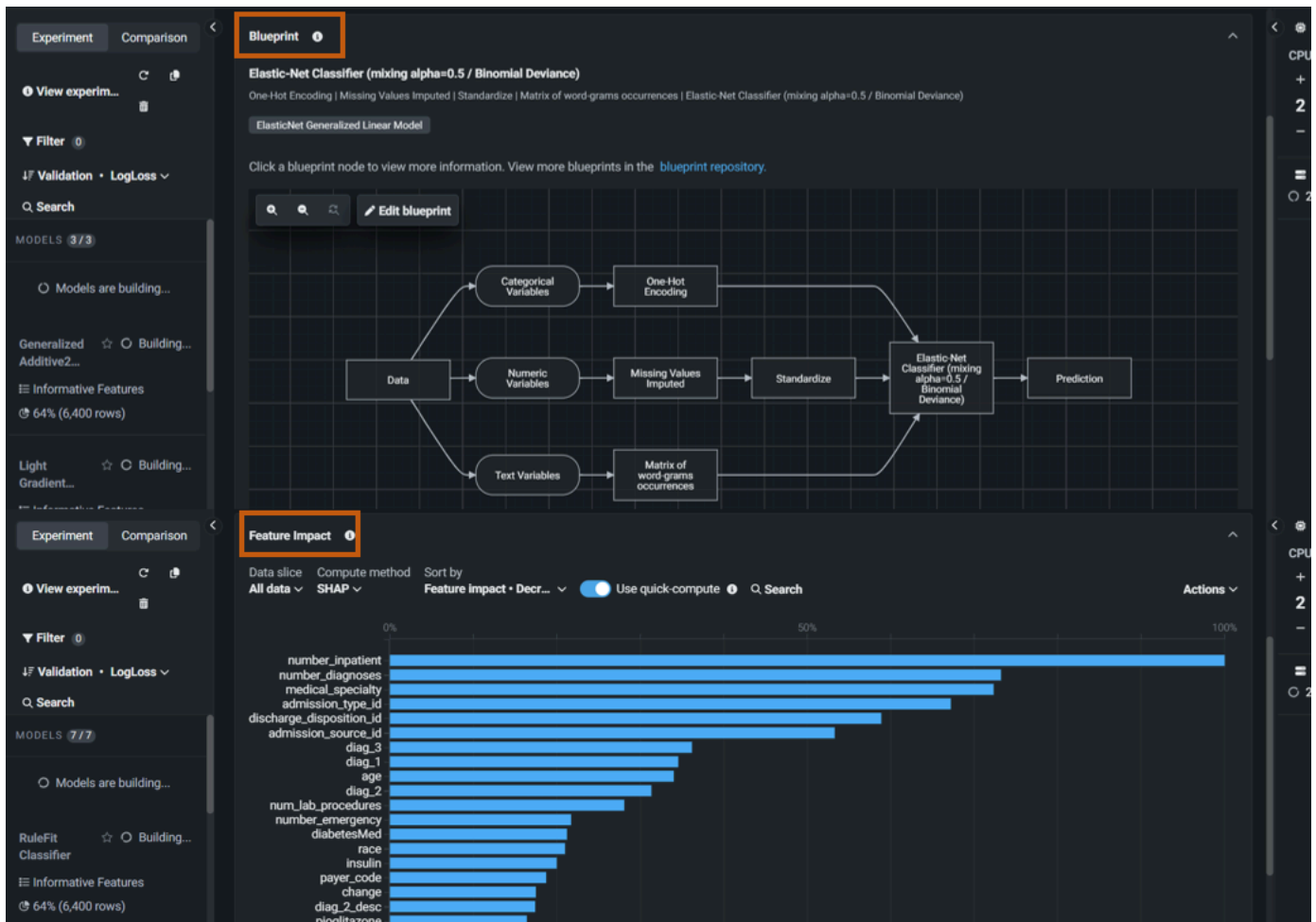
Step 19: once the modelling is complete, you can pick a model of your choice, and the DataRobot will show the **Model Overview**.

The screenshot shows the 'Model Overview' page for an 'Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)'. The left sidebar now shows '3/3' models, with the selected model highlighted in an orange box. The main area displays the model's performance and settings. The 'Training scores: LogLoss' table shows Validation (0.6089), Cross-validation (Score), and Holdout (0.6111). The 'Training settings' table shows Training feature list, Informative Features (64% (6,400 rows)), and Training sample size. Below these are sections for 'Blueprint', 'Feature Impact', 'Feature Effects', 'Individual Prediction Explanations', and 'ROC Curve'.

Training scores: LogLoss	
Validation	0.6089
Cross-validation	Score
Holdout	0.6111

Training settings	
Training feature list	Informative Features
Training sample size	64% (6,400 rows)

Step 20: You can explore various model overview components like **Blueprint**, **Feature Impact**, and so on.



Step 21: If you have test or unseen data, you can also make predictions by clicking **Make Predictions** under **Model actions**.

Model Overview
Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)

Training scores: LogLoss

Validation	Score
0.6089	
Cross-validation	0.6111
Holdout	

Training settings

- Training feature list
- Training sample size
- Informative Features: 64% (6,400 rows)

Make Predictions

Make new predictions

Prediction dataset
Prediction datasets cannot exceed 5.00 GB.

Drop file(s) here to upload or [Choose file](#)

Prediction options

- ☒ Include additional feature values in prediction
- ☐ Add all features ☐ Add specified features
- ☐ Include Prediction Explanations

Download recent predictions
Predictions are stored and available to download for 48 h

No recent predictions

Step 22: You can also click **Generate compliance report** and **download compliance report** for your use case.

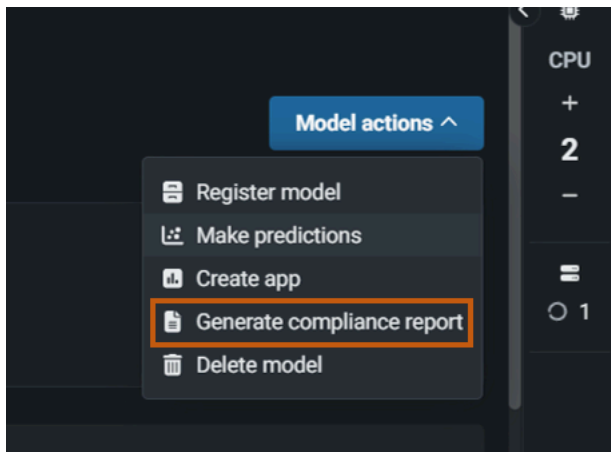


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 - 3.5 Model Interdependencies
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 - 4.2 Data Source Overview and Appropriateness
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Conclusion

In this lab, you have signed up in DataRobot, added a data set in a use case, and worked on data modelling.

Author(s)

[Dr. Pooja](#)



Skills Network